House Price Prediction

Project Overview

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property characteristics such as area, number of bedrooms, bathrooms, stories, parking availability, and other amenities. The Housing Prices Dataset was obtained from Kaggle and analyzed using Python, Pandas, Scikit-learn, Matplotlib, and Seaborn.

Data Preparation

The dataset was loaded and explored to understand its structure and identify potential issues. Missing values and duplicate records were checked and handled appropriately. Categorical features such as furnishing status, main road access, guest room availability, and air conditioning were converted into numerical format using one-hot encoding to make them suitable for machine learning algorithms.

Model Development

Two regression models were trained and evaluated:

1. Linear Regression
2. Random Forest Regressor

The dataset was split into training and testing sets using an 80:20 ratio. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Key Findings

The analysis revealed that property area was the most influential factor affecting house prices. Other important features included the number of bathrooms, number of stories, parking spaces, air conditioning, and location-related amenities. Houses with larger areas and better facilities generally had significantly higher market values.

Among the models tested, the Random Forest Regressor achieved better predictive performance than Linear Regression. This indicates that house prices depend on complex and non-linear relationships among features, which Random Forest can capture more effectively.

Visual Analysis

- Histogram of house prices showing the distribution of property values.
- Correlation heatmap highlighting relationships between features and house price.
- Actual vs. Predicted Price scatter plot demonstrating model performance.

Conclusion

The project successfully demonstrated how machine learning can be used to estimate house prices with reasonable accuracy. Property size, amenities, and infrastructure-related features were found to have the strongest impact on pricing. Real estate businesses can use predictive models like these to support property valuation, improve pricing strategies, and assist buyers and sellers in making informed decisions.

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